

Title

Strategic Integration of Ecological Principles in Multi-Component School Obesity Prevention: Implementation Patterns and Theoretical Alignment

Abstract

Thirty-two distinct intervention activities were identified across 15 schools targeting obesity prevention through modifications in nutrition and physical activity domains. Implementation pattern analysis revealed that strategies directed at institutional structures and personnel represented the largest category, followed by engagement of key administrative decision-makers from the education ministry. Fewer resources were allocated to direct parent involvement or individual student targeting. Examination of theoretical underpinning identified reciprocal determinism--the concept that environmental changes and behaviour change mutually reinforce one another--as the predominant framework across both intervention domains. Statistical testing showed no significant differential application of theoretical principles between nutrition-focused versus activity-focused components. The institutional emphasis of the intervention, combined with strong political partnership and theoretical grounding in behaviour-environment reciprocity, appears aligned with evidence-based implementation science recommendations. The relative underutilisation of parent and direct-student targeting represents an opportunity gap for future iterations. These data support the feasibility and acceptability of ecological multi-level approaches in resource-constrained school settings.